Random effects: WALHUS, AMEMIYA, Swamy-Arora and the nested model

Five ways to split a panel's error variance, written from one rule and tested against the tables of Baltagi (2005)

Tested against	Baltagi (2005), *Econometric Analysis of Panel Data*, 3rd ed.: Tables 2.1, 2.5, 2.6, 3.1, 3.6, 3.7 and 9.4. The investment data of Grunfeld (1958), the OECD gasoline data of Baltagi and Griffin (1983) and the public capital data of Munnell (1990) (PollisDatasets).
Missing in Julia	Wallace-Hussain (WALHUS), Amemiya (AMEMIYA), the two-way Swamy-Arora, and the estimators of the nested model (states within regions). Julia packages offer the one-way Swamy-Arora at most; R's plm has the rest.
Notebook	**Compose** › **New Methods** › **Browse Methods**: *WALHUS, AMEMIYA, Swamy-Arora and nested error components* (`panelre.ipynb`)
Result	288 of 298 printed values to every digit. R's plm agrees with Julia to its 7 printed significant digits wherever both use the same convention.

The problem

A random-effects panel regression is a GLS regression whose weights depend on the variances of the error components: an individual effect, perhaps a time effect or a nested effect, and the remainder. The estimators differ only in how they estimate those variances before GLS, and the choice moves the standard errors and sometimes the coefficients. A textbook that prints a row for each estimator cannot be followed in a language that has one of them.

One rule for all the estimators

Let $e = My$ be residuals, where M removes the regressors, so $e = Mu$ whatever the coefficients. A sum of squares $q = e'Ae$ has the exact expectation

$$\mathrm{E}(q) = \sum_c \sigma_c^2 \,\mathrm{tr}(M'AM\,Z_cZ_c'),$$

with Z_c the incidence matrix of component c. With as many sums of squares as components, equating each to its expectation gives the variances. This is how Baltagi (2005) defines the unbalanced estimators (Section 9.2.1, eqs. 9.16–9.21), and in balanced panels it gives the familiar formulas. The estimators are then only choices of residuals:

- **WALHUS**, Wallace and Hussain (1969): OLS residuals in every sum of squares.
- **AMEMIYA**, Amemiya (1971), and Wansbeek and Kapteyn (1989) when unbalanced: the residuals of the Within regression.
- **SWAR**, Swamy and Arora (1972): each sum of squares from its own regression, Within or Between.
- **HFB**, Henderson's method III, Henderson (1953), as in Fuller and Battese (1974): reductions in the sum of squares as the nested dummies are added.
- **MIVQUE**, Rao (1971), with a prior guess of the variances.

A negative variance is set to zero, as in Tables 3.1 and 3.7. Writing the rule once, rather than five sets of formulas, is what made the nested, unbalanced case as easy as the one-way case.

What the tests show

Table	Model and data	Agreement
2.1, p. 21	Grunfeld, one-way: all estimators	every printed digit
2.5, p. 24	Gasoline, one-way: WALHUS, AMEMIYA	every printed digit
2.5, p. 24	Gasoline, one-way: SWAR	variances agree; price $-.420$ vs $-.402$ (see below)
2.6, p. 25	Public capital, one-way	every printed digit
3.1, p. 43	Grunfeld, two-way	every printed digit
3.6, p. 46	Gasoline, two-way	every printed digit
3.7, p. 47	Public capital, two-way	every printed digit
9.4, p. 182	Public capital, nested: WH, WK, SA, MV1, MV2	every printed digit
9.4, p. 182	nested: HFB	six values in the last digit

The MLE and REML columns of Table 9.4 also agree to every printed digit; they are fitted with MixedModels.jl and check the data and the model. The EViews output Baltagi reprints for WALHUS in Table 2.2 is matched to every digit EViews prints: $\sigma_\mu = 87.358027$ and $\sigma_\nu = 53.745175$ here, against 87.35803 and 53.74518.

What the replication found

The book uses two conventions, and both had to be matched

The standard errors of the two-way tables (3.1, 3.6, 3.7) are scaled by the residual variance of the transformed regression, as EViews does; those of the one-way tables and Table 9.4 are not. With one convention for all, about a third of the standard errors differ in the last digit. And the sums of squares are equated to their exact expectations, not divided by the traces of eqs. (2.21)–(2.22): this gives AMEMIYA's remainder SWAR's divisor $N(T-1) - K$, which is why the two share $\sigma_\nu = 52.77$ in Table 2.1.

R's plm does not reproduce the book's WALHUS and AMEMIYA

plm divides by traces, so for one-way WALHUS it gives $\sigma_\mu = .078$ against Baltagi's .082 on the public capital data, and 75.43 against 87.36 on Grunfeld. In the two-way models it divides the between sums of squares by N and T rather than $N-1$ and $T-1$. Julia reproduces plm exactly with plm's conventions (`divisors = :trace`, `scale = true`), and the book exactly with the exact expectations: the two programs differ by a definition, not by an error.

A probable misprint in Table 2.5

For SWAR on the gasoline data, the variances agree ($\rho = .82$, $\sigma_\mu = .196$, $\sigma_\nu = .092$), but the price coefficient is $-.420$ here against $-.402$ in print. R's plm gives $-.4204$, as Julia does. The printed value looks like transposed digits.

Still open

- **One standard error in Table 2.5.** With the two-way scaling, Julia's SWAR standard errors are .059, .040 and .026 against .059, .042 and .026; plm gives .059, .040 and .0255. The .042 is not explained.
- **HFB in Table 9.4**, in the last digit: intercept 2.088 vs 2.084, log K .271 vs .272, log L .744 vs .743, its standard error .151 vs .150, σ_μ^2 .0027 vs .0029, σ_ν^2 .0046 vs .0044. The fitting-constants reductions follow Section 9.2.1; Baltagi, Song and Jung (2001) define the nested version, which may order them differently.
- **MV3.** The book does not give the MIVQUE priors. Priors $(0, 0, 1)$ and $(1, 1, 1)$ reproduce MV1 and MV2 exactly; no prior tried reproduces MV3 (the closest gives σ_μ^2 .0018 against .0017), so MV3 is not tested.

Further considerations

- **A note for users of panel software.** Two programs can both be "right" and still disagree in the second significant digit: the divisor and the scaling of the standard errors are conventions, and a replication has to name them. The notebook makes both explicit options, so either convention can be chosen deliberately.
- **Upstream.** The single quadratic-form rule is small and general enough to be offered to a Julia panel-data package, with these tables as its tests.
- **Next.** The Baltagi, Song and Jung (2001) nested HFB, to close the last-digit gap; the unbalanced two-way model; and the estimator of Hausman and Taylor (1981), also in Baltagi's chapters and also missing in Julia.

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